SAMPLE Examination in the School of Computer and Mathematical Sciences

STATS 7022	**SAMPLE EXAM** **Data Science PG**

Instructions:

- Attempt all questions.
- Start your answer to each question on a new page.
- Clearly label your answers by question and part.
- You must show working in your answers to every question. Full marks will not be awarded without supporting working.
- This is a closed book exam. You are only permitted access to the materials listed below.

Permitted Materials:

- A scientific calculator or a basic/four function calculator. Calculators with graphics, remote communication, or CAS capabilities are NOT permitted.
- One hard copy of an English language translation dictionary.

Duration:

Writing Time: 120 minutes

This exam has 4 questions for a total of 80 marks.

Question:	1	2	3	4	Total
Marks	20	20	20	20	80

DO NOT OPEN THIS BOOKLET OR COMMENCE WRITING UNTIL INSTRUCTED TO DO SO. STOP WRITING IMMEDIATELY WHEN INSTRUCTED.

20 marks

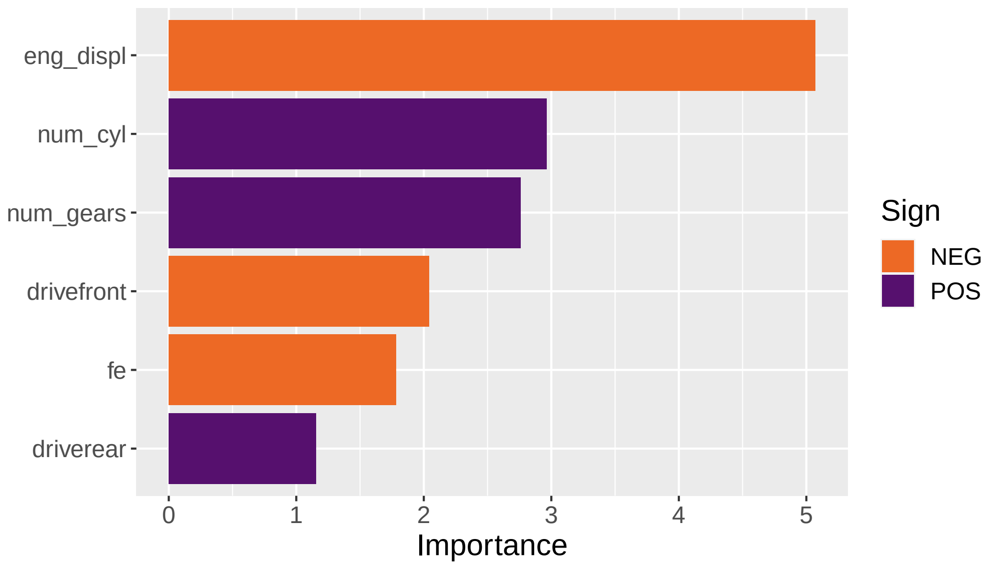
1. (a) A MARS model has been fitted to a random sample of the Hitters dataset. Using the following output, calculate the expected salary for a player with 63 runs batted in (`rbi`) and 9 home runs (`hm_run`). Please round your answer to the nearest 3 decimal places.

```
# Call: earth(formula=salary~rbi+hm_run, data=data,
#             keepxy=TRUE, nprune=~4)
#
# salary =
#   186.8413
#   + 31.19324 * pmax(0,     22 -    rbi)
#   + 12.62587 * pmax(0,    rbi -     22)
#   -  18.8807 * pmax(0, hm_run -     13)
#
# Selected 4 of 13 terms, and 2 of 2 predictors (nprune=4)
# Termination condition: Reached nk 21
# Importance: rbi, hm_run
# Number of terms at each degree of interaction: 1 3
# (additive model)
# GCV 160972.1    RSS 29497933    GRSq 0.1987167
# RSq 0.2470239
```

Solution:

$$\begin{aligned}\widehat{salary} &= 186.8413 + 31.19324 \times 0 + 12.62587 \times 41 - 18.8807 \times 0 \\ &= 704.502\end{aligned}$$

(b) A logistic regression model has been fitted to a random sample of the cars2011 dataset to predict whether a car's engine is "Turbocharged" (against the reference level of "Naturally Aspirated"). Using the variable importance plot below to answer the following questions:



(i) Which is the most important predictor in the model?

Solution: `eng_displ`.

(ii) Which is the most important positive predictor in the model?

Solution: `num_cyl`.

(iii) As `num_gears` increases, what do we expect will happen to the probability that our model predicts "Naturally Aspirated"?

Solution: The probability should decrease (`num_gears` is a positive predictor for "Turbocharged").

(iv) Is the importance value based on a z-statistic?

Solution: No, the importance value for a logistic regression model is based on a t-statistic.

(c) (i) Consider the following equation:

$$E\left[\left(Y_0 - \hat{f}(x_0)\right)^2\right] = \mathrm{var}\left(\hat{f}(x_0)\right) + \left(E\left[\hat{f}(x_0)\right] - f(x_0)\right)^2 + \mathrm{var}(\epsilon)$$

Which term in this equation is the bias?

Solution: $E\left[\hat{f}(x_0)\right] - f(x_0)$

(ii) Consider the following `tidymodels` workflow in `R`:

```
# == Workflow ==============================================
# Preprocessor: Recipe
# Model: logistic_reg()
#
# -- Preprocessor ------------------------------------------
# 3 Recipe Steps
#
# * step_dummy()
# * step_center()
# * step_BoxCox()
#
# -- Model -------------------------------------------------
# Logistic Regression Model Specification (classification)
#
# Computational engine: glm
```

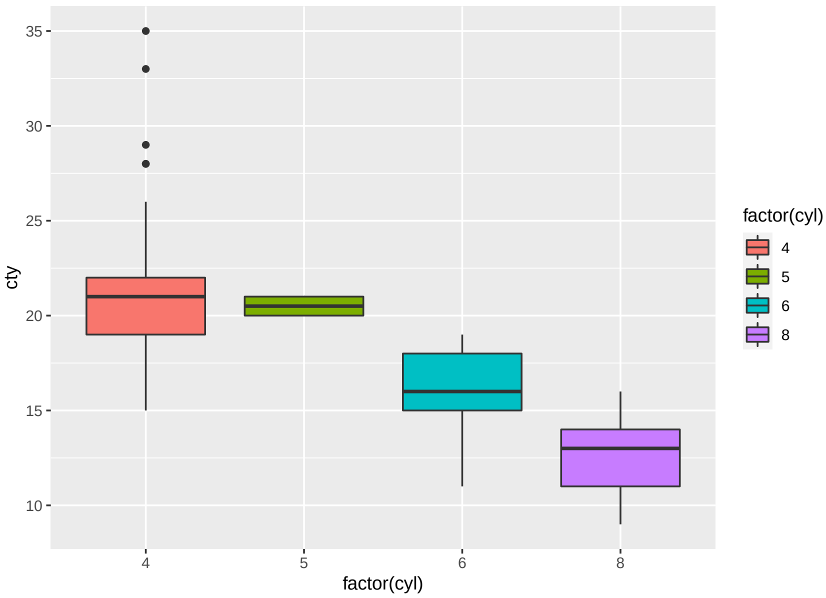
Which pre-processing steps does this workflow include?

- Centring
- Scaling
- Dealing with skewness

- Converting categorical variables to dummies
- Imputing categorical variables
- Imputing numeric variables

Solution:
Converting categorical variables to dummies (`step_dummy()`),
centring (`step_center()`), and
dealing with skewness (`step_BoxCox()`).

(iii) Consider the following plot:



Which if the following commands should be used to produce this plot in `R`?

- `ggplot(mpg, aes(displ, cty, col = factor(cyl))) + geom_point()`
- `ggplot(mpg, aes(cty, displ, col = factor(cyl))) + geom_point()`
- `ggplot(mpg, aes(factor(cyl), cty, fill = factor(cyl))) + geom_boxplot()`
- `ggplot(mpg, aes(factor(cyl), cty, col = factor(cyl))) + geom_boxplot()`

Solution:

```
ggplot(mpg,
       aes(factor(cyl),
           cty,
           fill = factor(cyl))) +
  geom_boxplot()
```

(iv) Consider following table in `R`:

```
# A tibble: 3 × 2
#   drv   `mean(hwy)`
#   <chr>       <dbl>
# 1 4            19.2
# 2 f            28.2
# 3 r            21
```

Which of the following commands could be used to produce this table?

- `mpg %>% group_by(cyl) %>% summarise(mean(cty))`
- `mpg %>% group_by(drv) %>% summarise(mean(cty))`
- `mpg %>% group_by(cyl) %>% summarise(mean(hwy))`
- `mpg %>% group_by(drv) %>% summarise(mean(hwy))`

Solution:

```
mpg %>% group_by(drv) %>% summarise(mean(hwy))
```

20 marks 2. Consider the linear regression model where the intercept term $\beta_0 = 0$:

$$y_i = \beta x_i + \epsilon_i$$

with ϵ_i are i.i.d. $N(0, \sigma^2)$.

(a) Derive $\hat{\beta}$, the least-squares estimate of β.

Solution:
We are trying to minimise

$$Q(\beta) = \sum_{i=1}^{n} (y_i - \beta x_i)^2$$

To find the value of β that minimises this, we differentiate with respect to β and set to zero.

$$\frac{dQ}{d\beta} = 2\sum_{i=1}^{n} (y_i - \beta x_i)(-x_i)$$

So

$$\begin{aligned}
0 &= 2\sum_{i=1}^{n} (y_i - \hat{\beta} x_i)(-x_i) \\
&= \sum_{i=1}^{n} (x_i y_i - \hat{\beta} x_i^2) \\
\sum_{i=1}^{n} x_i y_i &= \hat{\beta} \sum_{i=1}^{n} x_i^2 \\
\hat{\beta} &= \frac{\sum_{i=1}^{n} x_i y_i}{\sum_{i=1}^{n} x_i^2}
\end{aligned}$$

(b) Show that the least-squares estimate is unbiased.

Solution:
To show that $\hat{\beta}$ is unbiased, we need to show that $E[\hat{\beta}] = \beta$.

$$\begin{aligned}
E[\hat{\beta}] &= E\left[\frac{\sum_{i=1}^{n} x_i Y_i}{\sum_{i=1}^{n} x_i^2}\right] \\
&= \frac{\sum_{i=1}^{n} x_i E[Y_i]}{\sum_{i=1}^{n} x_i^2}
\end{aligned}$$

Note that

$$\begin{aligned}
E[Y_i] &= E[\beta x_i + \epsilon_i] \\
&= \beta x_i + 0
\end{aligned}$$

Thus

$$\begin{aligned} E[\hat{\beta}] &= \frac{\sum_{i=1}^n x_i \beta x_i}{\sum_{i=1}^n x_i^2} \\ &= \beta \frac{\sum_{i=1}^n x_i^2}{\sum_{i=1}^n x_i^2} \\ &= \beta \end{aligned}$$

(c) Derive the variance of $\hat{\beta}$, $\text{var}(\hat{\beta})$.

Solution:

$$\begin{aligned} \text{var}(\hat{\beta}) &= \text{var}\left(\frac{\sum_{i=1}^n x_i Y_i}{\sum_{i=1}^n x_i^2}\right) \\ &= \frac{1}{(\sum_{i=1}^n x_i^2)^2} \text{var}\left(\sum_{i=1}^n x_i Y_i\right) \text{ as } \text{var}(aX) = a^2 \text{var}(X) \\ &= \frac{1}{(\sum_{i=1}^n x_i^2)^2} \sum_{i=1}^n x_i^2 \text{var}(Y_i) \text{ as } Y_i \text{ are independent} \end{aligned}$$

Note that

$$\begin{aligned} \text{var}(Y_i) &= \text{var}(\beta x_i + \epsilon_i) \\ &= \text{var}(\epsilon_i) \\ &= \sigma^2 \end{aligned}$$

Thus

$$\begin{aligned} \frac{1}{(\sum_{i=1}^n x_i^2)^2} \sum_{i=1}^n x_i^2 \text{var}(Y_i) &= \frac{1}{(\sum_{i=1}^n x_i^2)^2} \sum_{i=1}^n x_i^2 \sigma^2 \\ &= \frac{\sigma^2}{\sum_{i=1}^n x_i^2} \end{aligned}$$

20 marks

3. There are four key skills in a data scientist's arsenal: analysis, mathematics, coding, and communication. In this question, we'll practice your communication skills. Whenever you are communicating, it's important to consider your audience. Different audiences will need to know different information, and be able to understand different levels of technical detail.

   Write a brief report explaining data cleaning and how to apply it in `tidymodels`. Your report should explain:

   - why data cleaning is important;
   - what data cleaning entails; and
   - how data cleaning is performed in `tidymodels`.

   Marks will be allocated for the quality and correctness of your writing, as well as how suitable your writing is for the intended audience.

   Solution:
   No solution is provided for this question, but submissions are to be marked according to the following criteria:

   - why data cleaning is important
   - what it entails
   - how it is performed in `tidymodels`
   - writing

   Writing will be assessed on: quality (is the writing smooth and easy-to-read), correctness (has correct spelling and grammar been used), and suitability (is the level of technical detail suitable for the intended audience).

20 marks

4. An analysis of the Titanic dataset is given on the following pages. Use the analysis to answer the following questions. Justify your answers by referring to the revelant section(s) of the analysis.

   (a) How many passengers are in the titanic dataset?

   Solution: 891 (see number of rows in tibble in Load data section).

   (b) Consider the following `R` code:

   ```
   titanic <-
     titanic %>%
     mutate(
       across(where(is.character), factor)
     )
   ```

   What does this code do?

   Solution: Converts all character variables to factors.

   (c) What predictors are in the final model?

   Solution: Passenger class; sex; age; number of siblings; number of parents or children (see select() in Clean data section (page 3)).

   (d) What tuning parameters are tuned in the analysis?

   Solution: The minimum number of obvservations required to split a node further, and tree depth (see hyperparameters set equal to tune() in Set up model section, or see Set up grid section).

   (e) What are the best tuning parameter values?

   Solution: `tree_depth` = 8; `min_n` = 2 (see show_best() table in Choose model section).

   (f) What is the AUC for the final model when fitted to the test dataset?

   Solution: 0.849 (see collect_metrics() in Choose model section).

   (g) What is the predicted probability of survival for a 3rd class 51 year old male travelling with his wife, but no parents or children?

   Solution: 0.109 (see final value calculated in Prediction section).

Titanic analysis

Jono Tuke

Last compiled on 05 October, 2021

Load data

```
data("titanic_train", package = "titanic")
titanic <- as_tibble(titanic_train)
titanic
```

```
## # A tibble: 891 x 12
##    PassengerId Survived Pclass Name   Sex     Age SibSp Parch Ticket  Fare Cabin
##          <int>    <int>  <int> <chr>  <chr> <dbl> <int> <int> <chr>  <dbl> <chr>
##  1           1        0      3 Braun~ male     22     1     0 A/5 2~  7.25 ""
##  2           2        1      1 Cumin~ fema~    38     1     0 PC 17~ 71.3  "C85"
##  3           3        1      3 Heikk~ fema~    26     0     0 STON/~  7.92 ""
##  4           4        1      1 Futre~ fema~    35     1     0 113803 53.1  "C12~
##  5           5        0      3 Allen~ male     35     0     0 373450  8.05 ""
##  6           6        0      3 Moran~ male     NA     0     0 330877  8.46 ""
##  7           7        0      1 McCar~ male     54     0     0 17463  51.9  "E46"
##  8           8        0      3 Palss~ male      2     3     1 349909 21.1  ""
##  9           9        1      3 Johns~ fema~    27     0     2 347742 11.1  ""
## 10          10        1      2 Nasse~ fema~    14     1     0 237736 30.1  ""
## # ... with 881 more rows, and 1 more variable: Embarked <chr>
```

Clean data

```
titanic <- titanic %>% clean_names()
titanic
```

```
## # A tibble: 891 x 12
##    passenger_id survived pclass name       sex     age sib_sp parch ticket  fare
##           <int>    <int>  <int> <chr>      <chr> <dbl>  <int> <int> <chr>  <dbl>
##  1            1        0      3 Braund, M~ male     22      1     0 A/5 2~  7.25
##  2            2        1      1 Cumings, ~ fema~    38      1     0 PC 17~ 71.3
##  3            3        1      3 Heikkinen~ fema~    26      0     0 STON/~  7.92
##  4            4        1      1 Futrelle,~ fema~    35      1     0 113803 53.1
##  5            5        0      3 Allen, Mr~ male     35      0     0 373450  8.05
##  6            6        0      3 Moran, Mr~ male     NA      0     0 330877  8.46
##  7            7        0      1 McCarthy,~ male     54      0     0 17463  51.9
##  8            8        0      3 Palsson, ~ male      2      3     1 349909 21.1
##  9            9        1      3 Johnson, ~ fema~    27      0     2 347742 11.1
## 10           10        1      2 Nasser, M~ fema~    14      1     0 237736 30.1
## # ... with 881 more rows, and 2 more variables: cabin <chr>, embarked <chr>
```

```r
titanic <-
  titanic %>%
  mutate(
    across(where(is.character), factor)
  )
titanic
```

```
## # A tibble: 891 x 12
##    passenger_id survived pclass name       sex     age sib_sp parch ticket  fare
##           <int>    <int>  <int> <fct>      <fct> <dbl>  <int> <int> <fct>  <dbl>
##  1            1        0      3 Braund, M~ male     22      1     0 A/5 2~  7.25
##  2            2        1      1 Cumings, ~ fema~    38      1     0 PC 17~ 71.3
##  3            3        1      3 Heikkinen~ fema~    26      0     0 STON/~  7.92
##  4            4        1      1 Futrelle,~ fema~    35      1     0 113803 53.1
##  5            5        0      3 Allen, Mr~ male     35      0     0 373450  8.05
##  6            6        0      3 Moran, Mr~ male     NA      0     0 330877  8.46
##  7            7        0      1 McCarthy,~ male     54      0     0 17463  51.9
##  8            8        0      3 Palsson, ~ male      2      3     1 349909 21.1
##  9            9        1      3 Johnson, ~ fema~    27      0     2 347742 11.1
## 10           10        1      2 Nasser, M~ fema~    14      1     0 237736 30.1
## # ... with 881 more rows, and 2 more variables: cabin <fct>, embarked <fct>
```

```r
titanic <-
  titanic %>%
  mutate(
    survived = factor(survived),
    pclass = factor(pclass)
  )
```

```r
titanic <-
  titanic %>%
  select(survived, pclass, sex, age, sib_sp, parch)
titanic
```

```
## # A tibble: 891 x 6
##    survived pclass sex      age sib_sp parch
##    <fct>    <fct>  <fct>  <dbl>  <int> <int>
##  1 0        3      male      22      1     0
##  2 1        1      female    38      1     0
##  3 1        3      female    26      0     0
##  4 1        1      female    35      1     0
##  5 0        3      male      35      0     0
##  6 0        3      male      NA      0     0
##  7 0        1      male      54      0     0
##  8 0        3      male       2      3     1
##  9 1        3      female    27      0     2
## 10 1        2      female    14      1     0
## # ... with 881 more rows
```

Split data

```r
titanic_split <- initial_split(titanic, strata = survived)
titanic_split
```

```
## <Analysis/Assess/Total>
```

```
## <667/224/891>
```

```
titanic_train <- training(titanic_split)
titanic_test <- testing(titanic_split)
```

```
titanic_cv <- vfold_cv(titanic_train, strata = survived)
titanic_cv
```

```
## #  10-fold cross-validation using stratification
## # A tibble: 10 x 2
##    splits           id
##    <list>           <chr>
##  1 <split [599/68]> Fold01
##  2 <split [600/67]> Fold02
##  3 <split [600/67]> Fold03
##  4 <split [600/67]> Fold04
##  5 <split [600/67]> Fold05
##  6 <split [600/67]> Fold06
##  7 <split [601/66]> Fold07
##  8 <split [601/66]> Fold08
##  9 <split [601/66]> Fold09
## 10 <split [601/66]> Fold10
```

Set up recipe

```
titanc_recipe <- recipe(survived ~ ., data = titanic)
```

Set up model

```
titanic_model <-
  decision_tree(
    mode = "classification",
    min_n = tune(),
    tree_depth = tune()
    ) %>%
  set_engine("rpart")
```

Set up workflow

```
titanic_wf <-
  workflow() %>%
  add_recipe(titanc_recipe) %>%
  add_model(titanic_model)
titanic_wf
```

```
## == Workflow ====================================================================
## Preprocessor: Recipe
## Model: decision_tree()
##
## -- Preprocessor ----------------------------------------------------------------
## 0 Recipe Steps
##
```

```
## -- Model -----------------------------------------------------------------------
## Decision Tree Model Specification (classification)
##
## Main Arguments:
##   tree_depth = tune()
##   min_n = tune()
##
## Computational engine: rpart
```

Set up grid

```
titanic_grid <- grid_regular(min_n(), tree_depth())
titanic_grid
```

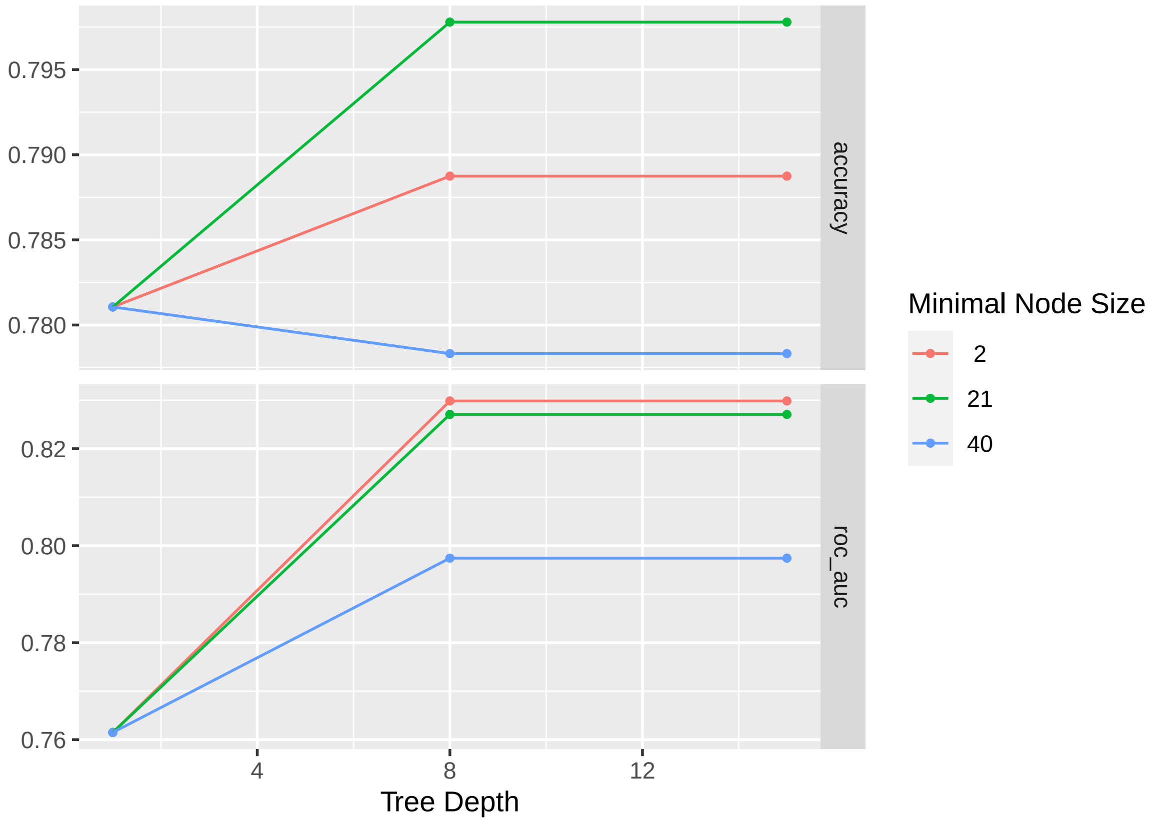
```
## # A tibble: 9 x 2
##   min_n tree_depth
##   <int>      <int>
## 1     2          1
## 2    21          1
## 3    40          1
## 4     2          8
## 5    21          8
## 6    40          8
## 7     2         15
## 8    21         15
## 9    40         15
```

Tune model

```
doParallel::registerDoParallel()
titanic_tune <- tune_grid(
  titanic_wf,
  resamples = titanic_cv,
  grid = titanic_grid,
  control = control_resamples(save_pred = TRUE)
)
```

Choose model

```
titanic_tune %>% autoplot()
```




```
show_best(titanic_tune, metric = "roc_auc")
```

```
## # A tibble: 5 x 8
##   tree_depth min_n .metric .estimator  mean     n std_err .config
##        <int> <int> <chr>   <chr>      <dbl> <int>   <dbl> <chr>
## 1          8     2 roc_auc binary     0.830    10  0.0204 Preprocessor1_Model4
## 2         15     2 roc_auc binary     0.830    10  0.0204 Preprocessor1_Model7
## 3          8    21 roc_auc binary     0.827    10  0.0214 Preprocessor1_Model5
## 4         15    21 roc_auc binary     0.827    10  0.0214 Preprocessor1_Model8
## 5          8    40 roc_auc binary     0.797    10  0.0205 Preprocessor1_Model6
```

```
titanic_wf <-
  titanic_wf %>%
  finalize_workflow(select_best(titanic_tune, metric = "roc_auc"))
```

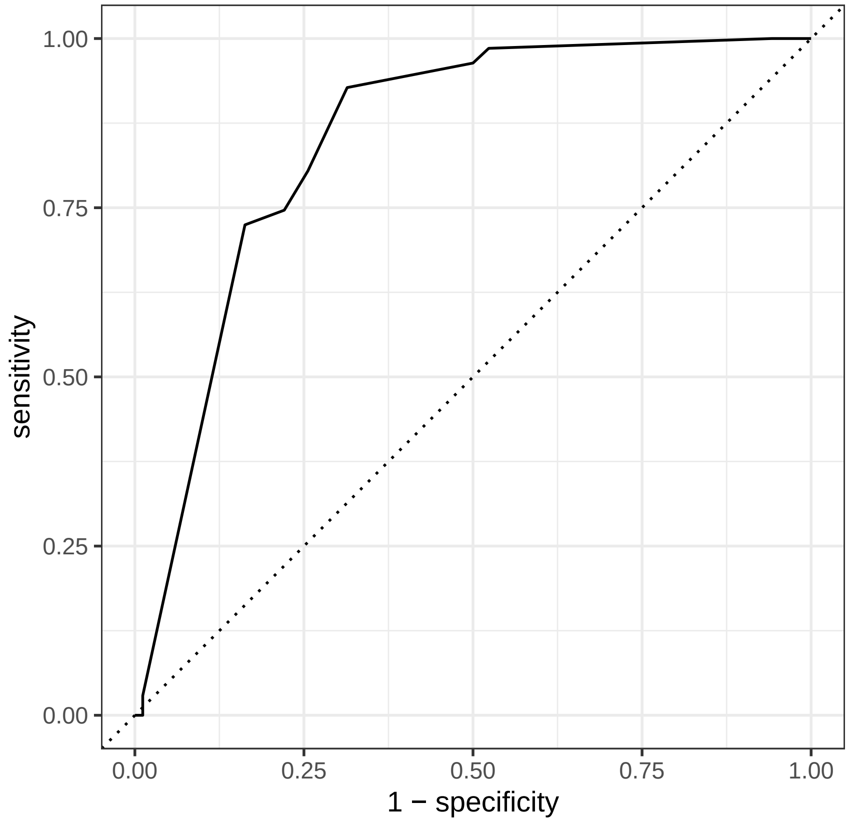
Quick last fit to see how works on test dataset.

```
titanic_LF <- titanic_wf %>% last_fit(titanic_split)
titanic_LF %>% collect_metrics()
```

```
## # A tibble: 2 x 4
##   .metric  .estimator .estimate .config
##   <chr>    <chr>          <dbl> <chr>
## 1 accuracy binary         0.835 Preprocessor1_Model1
## 2 roc_auc  binary         0.849 Preprocessor1_Model1
```

```
titanic_LF %>%
  collect_predictions() %>%
  roc_curve(truth = survived, estimate = .pred_0) %>%
```

```
  autoplot()
```



Prediction

```
newdata <- tibble(
  pclass = "3",
  sex = "male",
  age = 51,
  sib_sp = 1,
  parch = 0
)
titanic_LF %>%
  extract_fit_parsnip() %>%
  predict(new_data = newdata)
```

```
## # A tibble: 1 x 1
##   .pred_class
##   <fct>
## 1 0
```

```
titanic_LF %>%
  extract_fit_parsnip() %>%
  predict(new_data = newdata, type = "prob")
```

```
## # A tibble: 1 x 2
##   .pred_0 .pred_1
```

```
##      <dbl>   <dbl>
## 1    0.891   0.109
```